

1 Multiplication Rule for Counting

The multiplication rule is a fundamental principle in combinatorics. It states that if there are n_1 ways to do one task, n_2 ways to do another task, and so on, then the total number of ways to perform all tasks in sequence is the product of the number of ways for each task:

$$\text{Total ways} = n_1 \cdot n_2 \cdot n_3 \cdots n_k.$$

Example Explanation

Imagine you are choosing an outfit. You have:

- 3 shirts,
- 2 pairs of pants,
- 4 pairs of shoes.

The total number of outfits is given by:

$$3 \times 2 \times 4 = 24.$$

This means you have 24 different combinations of outfits.

Practice Problems

1. A password consists of 3 uppercase letters followed by 2 digits. How many unique passwords can be created?
2. A car license plate consists of 3 letters followed by 4 digits. How many unique license plates are possible?
3. A menu offers 2 appetizers, 3 main courses, and 2 desserts. How many meal combinations are possible?
4. A bookshelf has 5 shelves, and you can choose 1 book per shelf. If there are 4 books available for each shelf, how many ways can you arrange the books?
5. A survey offers 3 questions with 4 options each. How many possible ways can you answer all questions?

Solutions

1. $26^3 \times 10^2 = 17,576,000$.
2. $26^3 \times 10^4 = 17,576,000$.
3. $2 \times 3 \times 2 = 12$.
4. $4^5 = 1,024$.
5. $4^3 = 64$.

2 Permutations

A permutation is an arrangement of objects in a specific order. The number of permutations of n distinct objects taken r at a time is given by:

$$P(n, r) = \frac{n!}{(n-r)!}.$$

Key Points

- $n!$: "n factorial" is the product of all positive integers up to n . - Order matters in permutations.

Example Explanation

If you have 5 books and want to arrange 3 of them on a shelf, the number of arrangements is:

$$P(5, 3) = \frac{5!}{(5-3)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 60.$$

This means there are 60 different arrangements of 3 books.

Practice Problems

1. How many ways can 4 people be arranged in a line?
2. From a group of 6 runners, how many ways can you award 1st, 2nd, and 3rd place?
3. A bookshelf has 7 books. How many ways can you arrange 4 of them?
4. A code uses 5 digits, and no digit repeats. How many codes are possible?
5. How many ways can 5 students sit in 3 chairs?

Solutions

1. $4! = 24$.
2. $P(6, 3) = \frac{6!}{(6-3)!} = 6 \times 5 \times 4 = 120$.
3. $P(7, 4) = \frac{7!}{(7-4)!} = 7 \times 6 \times 5 \times 4 = 840$.
4. $P(10, 5) = \frac{10!}{(10-5)!} = 10 \times 9 \times 8 \times 7 \times 6 = 30,240$.
5. $P(5, 3) = \frac{5!}{(5-3)!} = 5 \times 4 \times 3 = 60$.

Probability

Probability is the measure of the likelihood of an event occurring. It is calculated as the ratio of favorable outcomes to the total number of possible outcomes:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}.$$

Key Concepts

- **Sample Space (S):** The set of all possible outcomes. - **Event (A):** A subset of the sample space. - **Complementary Events:** $P(A^c) = 1 - P(A)$. - **Addition Rule:** For events A and B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. - **Multiplication Rule:** For independent events A and B, $P(A \cap B) = P(A) \cdot P(B)$.

Example Explanation

If you roll a fair six-sided die, the probability of rolling an even number is:

$$P(\text{even}) = \frac{\text{Number of even outcomes}}{\text{Total outcomes}} = \frac{3}{6} = \frac{1}{2}.$$

Practice Problems

1. A coin is flipped twice. What is the probability of getting at least one head?
2. A bag contains 3 red balls and 5 blue balls. What is the probability of drawing a red ball?
3. A six-sided die is rolled. What is the probability of rolling a number greater than 4?
4. Two cards are drawn from a standard deck of 52 cards without replacement. What is the probability that both are aces?
5. A spinner with 8 equal sections is spun. What is the probability of landing on an odd-numbered section?

Solutions

1. $P(\text{at least one head}) = 1 - P(\text{no heads}) = 1 - \frac{1}{4} = \frac{3}{4}$.
2. $P(\text{red ball}) = \frac{3}{3+5} = \frac{3}{8}$.
3. $P(\text{greater than 4}) = \frac{2}{6} = \frac{1}{3}$.
4. $P(\text{both aces}) = \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$.
5. $P(\text{odd section}) = \frac{4}{8} = \frac{1}{2}$.

Expected Value

The expected value (EV) of a random variable is a measure of its central tendency, calculated as the weighted average of all possible values, weighted by their probabilities:

$$E(X) = \sum_i x_i P(x_i).$$

Key Concepts

- **Discrete Random Variables:** $E(X) = \sum x_i P(x_i)$. - **Continuous Random Variables:** $E(X) = \int_{-\infty}^{\infty} x f(x) dx$. - **Linearity of Expectation:** $E(aX + b) = aE(X) + b$.

Example Explanation

Consider a game where you win \$10 if you roll a 6 on a die, and lose \$2 otherwise. The expected value of your winnings is:

$$E(X) = 10 \cdot \frac{1}{6} + (-2) \cdot \frac{5}{6} = \frac{10}{6} - \frac{10}{6} = 0.$$

This means the game is fair.

Practice Problems

1. A lottery ticket costs \$2 and has a 1 in 100 chance of winning \$150. What is the expected value of buying a ticket?
2. A die is rolled. What is the expected value of the number rolled?
3. In a game, you win \$5 with probability $\frac{1}{3}$ and lose \$3 with probability $\frac{2}{3}$. What is the expected value?
4. A spinner has values 1, 2, 3, and 4, each equally likely. What is the expected value of the result?
5. An investment has a 50% chance of gaining \$100 and a 50% chance of losing \$50. What is the expected value?

Solutions

1. $E(X) = 150 \cdot \frac{1}{100} - 2 = 1.5 - 2 = -0.5$.
2. $E(X) = \frac{1+2+3+4+5+6}{6} = 3.5$.
3. $E(X) = 5 \cdot \frac{1}{3} - 3 \cdot \frac{2}{3} = \frac{5}{3} - 2 = -\frac{1}{3}$.
4. $E(X) = \frac{1+2+3+4}{4} = 2.5$.
5. $E(X) = 100 \cdot 0.5 - 50 \cdot 0.5 = 50 - 25 = 25$.